

MySQL Day-1

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- What is Data?
Raw facts or figures without context
Eg: 111, 20000, raj, ram, vizag
- What is field?
Smallest unit of data in database(Column/attribute)
Eg: ename, esalary, eaddress, eid
- What is record?
Collection of related fields(row/tuple)
Eg: (101, 'Sunny', 50000)
- What is database?
Organized collection of related records stored together
Eg: an employee database containing all employee records.
- Client-Server Architecture
 - Client: The role of the Client in Client-Server architecture is to send request to the server and to get response from the server.
 - Server: Server is a software. The role of the server in Client-Server architecture is to take request from the Clients. If the request is valid request it will generate some response and we can handover that response to Client. If the Client request is not a valid request it will raise some error status codes.
 - Protocol: It has a set of rules and regulations. The role of the protocol in Client-Server architecture is to send request from Client to Server.
- Identifiers
SQL identifiers are names used to identify database objects such as.
Database names, Table names, Column names, view names, index names, constraints, triggers, stored procedures.

Rules for Identifiers:
 - We can use a-z, A-Z, 0-9, _, #.
 - Identifiers should not start with digits.
 - Spaces are not considered to declare identifiers.
 - Reserved words are not considered to declare identifiers.
 - Max length to declare identifiers is 15.
- SQL = Structured Query Language
SQL is a Standardized programming language specially designed for managing and manipulating relational databases. It provides a declarative syntax that allows users to define what data they want to retrieve, insert, update or delete without specifying exactly how database should execute these operations.

➤ SQL Commands

○ DDL = Data Definition Language

Defines and modifies the structure of database objects (Tables, Schemas, Indexes)

Commands:

Create: Creates a new database objects such as tables, databases, views, indexes, or triggers, etc.

Alter :

Drop: Permanently deletes an existing database objects (table, database, index, views) along with its data and structure.

Truncate

Rename

○ DML = Data Manipulation Language

Manages data stored within database objects (insert, update, delete)

○ DCL = Data Control Language

Manages permissions and access control on database objects.

○ TCL = Transaction Control Language

Manages Transactions, to maintain data integrity.

○ DQL = Data Query Language

Used to fetch/retrieve data from the database.

➤ Create a Database/Schema

Syntax: CREATE DATABASE database_name;

Or

CREATE SCHEMA Schema_name;

➤ Create a table

Syntax: CREATE TABLE table_name(
Column_name datatype(size) constraint,
Column_name datatype(size) constraint,
Column_name datatype(size) constraint,
....
Column_name datatype(size) constraint);

➤ Drop a Database/Schema

Syntax: DROP DATABASE database_name;

Or

DROP SCHEMA Schema_name;

➤ Drop a Table

Syntax: DROP TABLE table_name;